AIT304	ROBOTICS AND INTELLIGENT SYSTEM	Category	L	T	P	Credit	Year of Introduction
		PCC	3	1	0	4	2022

Preamble: This course enables the learners to understand the fundamental concepts and algorithms in Robotics and Intelligent systems. The course covers the standard hardware and kinematic concepts for robot design. Standard algorithms for localization, mapping, path planning, navigation and obstacle avoidance, to incorporate intelligence in robots are included in the course. This course helps the students to design robots with intelligence in a real world environment.

Prerequisite: Basic understanding of probability theory, linear algebra, machine learning, artificial intelligence

Course Outcomes: After the completion of the course the student will be able to

CO1	Understand the concepts of manipulator and mobile robotics. **(Cognitive Knowledge Level: Understand)**
CO2	Choose the suitable sensors, actuators and control for robot design. **(Cognitive Knowledge Level: Apply)**
CO3	Developing kinematic model of mobile robot and understand robotic vision intelligence. **(Cognitive Knowledge Level: Apply)**
CO4	Discover the localization and mapping methods in robotics. **(Cognitive Knowledge Level: Apply)**
CO5	Plan the path and navigation of robot by applying artificial intelligence algorithm. **(Cognitive Knowledge Level: Apply)**

Mapping of course outcomes with program outcomes

	PO 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO1 0	PO11	PO1 2
CO1	✔				✔	✔						✔
CO2	✔	✔			✔							✔
CO3	✔	✔		✔	✔	✔						✔
CO4	✔			✔	✔	✔						✔
CO5	✔			✔	✔	✔						✔

Abstract POs defined by National Board of Accreditation			
PO#	**Broad PO**	**PO#**	**Broad PO**
PO1	Engineering Knowledge	**PO7**	Environment and Sustainability
PO2	Problem Analysis	**PO8**	Ethics
PO3	Design/Development of solutions	**PO9**	Individual and team work
PO4	Conduct investigations of complex problems	**PO10**	Communication
PO5	Modern tool usage	**PO11**	Project Management and Finance
PO6	The Engineer and Society	**PO12**	Life long learning

Assessment Pattern

Bloom's Category	Continuous Assessment Tests		End Semester Examination Marks (%)
	Test 1 (%)	Test 2 (%)	
Remember	30	30	30
Understand	30	30	30
Apply	40	40	40
Analyze			
Evaluate			
Create			

Mark Distribution

Total Marks	CIE Marks	ESE Marks	ESE Duration
150	50	100	3

Continuous Internal Evaluation Pattern:

Attendance	**10 marks**
Continuous Assessment Tests(Average of Internal Tests 1 & 2)	**25 marks**
Continuous Assessment Assignment	**15 marks**

Internal Examination Pattern

Each of the two internal examinations has to be conducted out of 50 marks. First series test shall be preferably conducted after completing the first half of the syllabus and the second series test shall be preferably conducted after completing remaining part of the syllabus. There will be two parts: Part A and Part B. Part A contains 5 questions (preferably, 2 questions each from the completed modules and 1 question from the partly completed module), having 3 marks for each question adding up to 15 marks for part A. Students should answer all questions from Part A. Part B contains 7 questions (preferably, 3 questions each from the completed modules and 1 question from the partly completed module), each with 7 marks. Out of the 7 questions, a student should answer any 5.

End Semester Examination Pattern: There will be two parts; Part A and Part B. Part A contains 10 questions with 2 questions from each module, having 3 marks for each question. Students should answer all questions. Part B contains 2 full questions from each module of which student should answer any one. Each question can have a maximum 2 subdivisions and carries 14 marks.

SYLLABUS

Module – 1 (Introduction to robotics)

Introduction to robotics – Degrees of freedom, Robot types- Manipulators- Anatomy of a robotic manipulator-links, joints, actuators, sensors, controllers. Robot configurations-PPP, RPP, RRP, RRR. Mobile robots- wheeled, legged, aerial robots, underwater robots, surface water robots . Dynamic characteristics- speed of motion, load carrying capacity & speed of response. Introduction to End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and Passive grippers. Ethics in robotics - 3 laws - applications of robots.

Module - 2(Sensors, Actuators and Control)

Sensor classification- touch, force, proximity, vision sensors. Internal sensors-Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors-contact type, non contact type; Digital Camera - CCD camera - CMOS camera - Omnidirectional cameras
Sensor characteristics. Actuators - DC Motors - H-Bridge - Pulse Width Modulation - Stepper Motors – Servos, Hydraulic & pneumatic actuators. Control - On-Off Control - PID Control - Velocity Control and Position Control

Module – 3 (Robotic vision & Kinematics)

Robotic Vision: Sensing, Pre-processing, Segmentation, Description, Recognition, Interpretation, Feature extraction -Camera sensor hardware interfacing. Representation of Transformations - Representation of a Pure Translation - - Pure Rotation about an Axis - Combined Transformations - Transformations Relative to the Rotating Frame.

Basic understanding of Differential-Drive Wheeled Mobile Robot, Car-Like Wheeled Mobile Robot. Kinematic model of a differential drive and a steered mobile robot, Degree of freedom and manoeuvrability, Degree of steerability, Degree of mobility - different wheel configurations, holonomic and nonholonomic robots. Omnidirectional Wheeled Mobile Robots.

Module - 4 (Localization and Mapping)

Position and Orientation - Representing robot position. Basics of reactive navigation; Robot Localization, Challenges in localization - An error model for odometric position estimation

Map Representation - Continuous representations - Decomposition strategies - Current challenges in map representation. Probabilistic map-based localization (only Kalman method), Autonomous map building, Simultaneous localization and mapping (SLAM) - Mathematical definition of SLAM - Visual SLAM with a single camera - Graph-based SLAM - Particle filter SLAM - Open challenges in SLAM

Module - 5 (Path Planning and Navigation)

Path Planning- Graph search, deterministic graph search - , breadth first search - depth first search- Dijkstra' s algorithm, A*, D* algorithms, Potential field based path planning. Obstacle avoidance - Bug algorithm - Vector Field Histogram - Dynamic window approaches. Navigation Architectures - Modularity for code reuse and sharing - Control localization - Techniques for decomposition. Alternatives for navigation - Neural networks - Processing the image - Training the neural network for navigation - Convolutional neural network robot control implementation

Text Books

1. R Siegwart, IR Nourbakhsh, D Scaramuzza, Introduction to Autonomous Mobile Robots ,, MIT Press, USA, 2011
2. Thomas Bräunl - Embedded Robotics,Mobile Robot Design and Applications with Embedded Systems-Springer (2006)
3. S.G. Tzafestas - Introduction to Mobile Robot Control-Elsevier (2014)
4. Francis X. Govers - Artificial Intelligence for Robotics-Packt Publishing (2018)
5. Saeed B. Niku - Introduction to Robotics_ Analysis, Control, Applications

Reference Books

1. John J. Craig, Introduction to Robotics, Pearson Education Inc., Asia, 3rd Edition, 2005
2. S. K. Saha, Introduction to Robotics 2e, TATA McGraw Hills Education (2014)
3. Peter Corke - Robotics, Vision and Control_ Fundamental Algorithms in MATLAB® - Springer-Verlag Berlin Heidelberg (2021)

Course Level Assessment Questions

Course Outcome1 (CO1):

1. Categorise the various types of Grippers used in robot manipulators.
2. Differentiate between active and passive grippers.
3. Explain speed of motion and load carrying capacity of a mobile robot.
4. You wish to build a dynamically stable robot with a single wheel only. For each of the four basic wheel types, explain whether or not it may be used for such a robot.

Course Outcome 2(CO2):

1. Categorise the sensors used in robotics
2. Explain any four characteristics of a sensor
3. Illustrate the sensor performance measuring parameters
4. Suggest any two mechanism to realise 360^o Camera

Course Outcome 3(CO3):

1. Determine the degrees of mobility, steerability, and maneuverability for each of the following: (a) bicycle; (b) dynamically balanced robot with a single spherical wheel (c) automobile.
2. A frame F was rotated about the y-axis 90^o, followed by a rotation about the o-axis of 30^o, followed by a translation of 5 units along the n-axis, and finally, a translation of 4 units along the x-axis. Find the total transformation matrix.
3. Explain the camera sensor hardware interfacing.
4. What is an omni directional robot? Explain two configurations to set up an omni directional robot.

Course Outcome 4(CO4): .

1. Explain the challenges of localization
2. How Kalman method can be used in localization of mobile robots
3. What are the Decomposition strategies in map representation
4. How Visual SLAM can be performed with a single camera

Course Outcome 5(CO5):

1. Explain Dijkstra's algorithm with a suitable example.
2. Identify the steps of Generic temporal decomposition of a navigation architecture.
3. What is meant by control decomposition? Explain two types of control decomposition.
4. Why does SLAM work better with wheel odometer data available?

5. In the Floor Finder algorithm, what does the Gaussian blur function does to improve the results?

Model Question Paper

QP CODE:

Reg No: _______________

Name: _________________ **PAGES : 4**

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SIXTH SEMESTER B.TECH DEGREE EXAMINATION, MONTH & YEAR

Course Code: AIT304

Course Name: ROBOTICS AND INTELLIGENT SYSTEM

Max. Marks : 100 **Duration: 3 Hours**

PART A

Answer All Questions. Each Question Carries 3 Marks

1. What do you mean by degrees of freedom? How many degrees of freedom are required for a drone to achieve any position in 3D space? And how many more DOF required for achieving any orientation as well.

2. Explain how leg configuration affects the stability of mobile robot.

3. Explain Dynamic range, Linearity and Resolution of a Sensor.

4. Explain the working of a Mechanical accelerometer with a block diagram

5. Differentiate between holonomic and nonholonomic robots.

6. What is the significance of differential drive in mobile robot?

7. How will you represent the position and orientation of a wheeled mobile robot?

8. Identify the 2 mobile robot localization problems.

9. Explain the Bug algorithm for obstacle avoidance. .

10. What is Voronoi diagram method and its advantages? **(10x3=30)**

Part B

(Answer any one question from each module. Each question carries 14 Marks)

11. (a) Explain the general features of wheeled, legged and aerial robots. **(9)**

(b) Explain the anatomy of a robotic manipulator with a neat diagram. **(5)**

OR

12. (a) Briefly explain the dynamic characteristics of robots. **(9)**

(b) Assume an object of mass 140 kg is to be lifted up with an acceleration of 10 m/s2. Calculate the gripper force required for the operation, if coefficient of friction between contact surfaces is 0.2, number of fingers in gripper is 2 and acceleration due to gravity is 9.8 m/s2 **(5)**

13. (a) Explain the working of an Optical Encoder. **(5)**

(b) A mobile robot is designed for unidirectional motion with constant velocity. Illustrate the mechanism to make the robot move in forward and reverse direction with variable speed. Support with necessary diagrams **(9)**

OR

14. (a) Compare and contrast the working of CCD and CMOS camera **(9)**

(b) Illustrate the significance of the PID controller with a neat block diagram **(5)**

15. (a) Outline the seven stages of robot vision. **(14)**

OR

16. (a) Derive the kinematic model of a differential drive mobile robot. **(7)**

(b) A frame B was rotated about the x-axis 90^o, then it was translated about the current a-axis 3 inches before it was rotated about the z-axis 90^o. Finally, it was translated about the current o–axis 5 inches. **(7)**

(a) Write an equation that describes the motions.

(b) Find the final location of a point p(1,5,4)T attached to the frame relative to the reference frame.

17. (a) Derive error model for odometric position estimation **(8)**

(b) Illustrate the SLAM problem with suitable diagrams **(6)**

OR

18. (a) Compare and Contrast graph based and particle SLAM **(8)**

(b) Describe the concept of mobile robot localization with suitable Block diagrams **(6)**

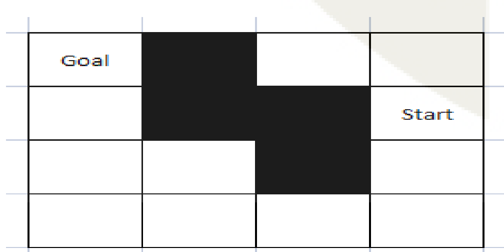
19. (a) Compare and contrast local and global Dynamic window approaches in obstacle avoidance. **(7)**

(b) Explain the concepts of floor finding Algorithm **(7)**

OR

20. (a) Illustrate the Incorporation of Neural network approach in Robot navigation? List its advantages **(6)**

(b) Make the robot to run from start position to goal position in the Following diagram using A* Algorithm **(8)**



TEACHING PLAN

No	Contents	No. of Lecture Hours (45 hrs)
Module-1 (Introduction to robotics) (8 hours)		
1.1	Introduction to robotics – Degrees of freedom - Robot types	1 hour
1.2	Manipulators- Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller	1 hour
1.3	Robot configurations-PPP, RPP, RRP, RRR- Mobile robots- wheeled	1 hour
1.4	Legged robots, Aerial robots, underwater robots, surface water robots -	1 hour
1.5	Dynamic characteristics of robot- speed of motion, load carrying capacity & speed of response	1 hour
1.6	Introduction to End effectors - mechanical grippers, special tools, Magnetic grippers	1 hour
1.7	Vacuum grippers, adhesive grippers, Active and Passive grippers	1 hour
1.8	Ethics in robotics - 3 laws - applications of robots	1 hour
Module-2 (Sensors, Actuators and Control) (9 hours)		
2.1	Sensor classification- touch, force, proximity, vision sensors.	1 hour
2.2	Internal sensors-Position sensors, velocity sensors	1 hour
2.3	Acceleration sensors, Force sensors;	1 hour
2.4	External sensors-contact type, non-contact type	1 hour

2.5	Digital Camera - CCD camera - CMOS camera	1 hour
2.6	Omnidirectional cameras - Sensor characteristics	1 hour
2.7	Actuators - DC Motors - H-Bridge - Pulse Width Modulation	1 hour
2.8	Stepper Motors – Servos - Control - On-Off Control	1 hour
2.9	PID Control - Velocity Control and Position Control	1 hour
Module-3 (Robotic vision & Kinematics) (9 hours)		
3.1	Robot Vision: Sensing, Pre-processing, Segmentation, Description	1 hour
3.2	Recognition, Interpretation, Feature extraction -Camera sensor hardware interfacing	1 hour
3.3	Representation of Transformations - Representation of a Pure Translation - Pure Rotation about an Axis	1 hour
3.4	Combined Transformations - Transformations Relative to the Rotating Frame	1 hour
3.5	Basic understanding of Differential Drive Wheeled Mobile Robot - Car Like Wheeled Mobile Robot	1 hour
3.6	Kinematic model of a differential drive and a steered mobile robot.	1 hour
3.7	Degree of freedom and manoeuvrability, Degree of steerability	1 hour
3.8	Degree of mobility, Different wheel configurations	1 hour
3.9	Holonomic and Nonholonomic robots, Omnidirectional Wheeled Mobile Robots	1 hour
Module-4 (Localization and Mapping) (9 hours)		

4.1	Position and Orientation - Representing robot position, Basics of reactive navigation	1 hour
4.2	Robot Localization, Challenges in localization	1 hour
4.3	An error model for odometric position estimation	1 hour
4.4	Map Representation - Continuous representations - Decomposition strategies	1 hour
4.5	Current challenges in map representation, Probabilistic map-based localization (only Kalman method)	1 hour
4.6	Probabilistic map-based localization (only Kalman method)	1 hour
4.7	Autonomous map building, Simultaneous localization and mapping (SLAM) - Mathematical definition of SLAM	1 hour
4.8	Visual SLAM with a single camera - Graph-based SLAM	1 hour
4.9	Particle filter SLAM - Open challenges in SLAM	1 hour
Module-5 (Path Planning and Navigation) (10 hours)		
5.1	Path Planning- Graph search	1 hour
5.2	Deterministic graph search - breadth first search - depth first search- Dijkstra's algorithm	1 hour
5.3	A*, D* algorithms, Potential field based path planning	1.5 hour
5.4	Obstacle avoidance - Bug algorithm - Vector Field Histogram - Dynamic window approaches	1.5 hour

5.5	Navigation Architectures - Modularity for code reuse and sharing - Control localization - Techniques for decomposition	1 hour
5.6	Alternatives for navigation - Neural networks	1 hour
5.7	Processing the image - Training the neural network for navigation	1.5 hour
5.8	Training the neural network for navigation - Convolutional neural network robot control implementation	1.5 hour

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